PROMYS Solutions by Yash Tanwar

Q.1

Start with a positive integer, then choose a negative integer. We'll use these two numbers to generate a sequence using the following rule: create the next term in the sequence by adding the previous two. For example, if we started with 6 and −5, we would get the sequence
6, −5, 1, −4 alternating part , −3, −7, −10, −17, −27, . . .
which starts with 4 elements that alternate sign before the terms are all negative. If we started with 3 and −2, we would get the sequence
3, −2, 1, −1 alternating part , 0, −1, −1, −2, −3, . . .
which also starts with 4 elements that alternate sign before the terms are all non-positive (we don't count 0 in the alternating part).
(a) Can you find a sequence of this type that starts with 5 elements that alternate sign? With 10 elements that alternate sign? Can you find a sequence with any number of elements that alternate sign?
(b) Given a particular starting integer, what negative number should you choose to make the alternating part of the sequence as long as possible? For example, if your sequence started with 8, what negative number would give the longest alternating part? What if you started with 10? With n?

Solution: -

- (8,-5) shows 5 elements with alternate signs.
- (89,-55) shows 10 elements with alternate signs.
- The number of alternating terms depends on the value of x and y.

The sequence follows a Fibonacci-like growth pattern, where each term is the sum of the two preceding terms, just like the Fibonacci sequence.
The sequence stabilizes when the alternating terms become negative or non-positive, much like how the Fibonacci sequence grows exponentially.

The alternating sequence typically does not exceed $\frac{x}{2}$, as after this length, the terms tend to stabilize and stop alternating in sign.

k represents the difference between the maximum possible alternating part length $\frac{x}{2}$ and the actual alternating part length.
$k = \frac{x}{2} - (actual\ alternating\ part\ length)$

For higher values of x, we observe that the value of k tends to increase or stabilize around 0.
Larger values of x lead to longer alternating parts (closer to $\frac{x}{2}$).

Prediction of k Based on x:

Based on observations, we can predict k values for given x using the following general trends:

- For smaller x, k is typically larger, as the alternating part tends to be shorter.
- For larger x, k becomes smaller, reflecting that the alternating part length approaches $\frac{x}{2}$.

x	Max. Alternating	Predicted k	Reasoning
6	3	0	Alternates for 3 terms before stabilizing.
7	4	-1	Alternating part length is 4, but $\frac{7}{2}$=3.5
8	5	0	Alternates for 5 terms before stabilizing.
9	4	0	Alternates for 4 terms, and the sequence stabilizes
10	4	1	Alternates for 4 terms, but $\frac{10}{2}$=5

As x increases k also increases, thus there is a complex relation between k and x. With this we would be able to find the negative integer required to stabilize with maximum alternating elements.

I used python to generate negative terms and alternating element numbers to analyse the value and growth of k.

Code:

```python
def generate_sequence(x, y, max_length=100):
    sequence = [x, y]
    while len(sequence) < max_length:
        sequence.append(sequence[-1] + sequence[-2])
    return sequence

def find_alternating_part(sequence):
    alternating_part = []
    for i in range(1, len(sequence)):
        if (sequence[i] > 0 and sequence[i-1] < 0) or (sequence[i] < 0 and sequence[i-1] > 0):
            alternating_part.append(sequence[i-1])
        else:
            break
    return alternating_part

def main():
    for x in range(1, 101):
        max_alternating_length, best_y = 0, None
        for y in range(1, 101):
            sequence = generate_sequence(x, -y)
            alternating_part = find_alternating_part(sequence)
            if len(alternating_part) > max_alternating_length:
                max_alternating_length, best_y = len(alternating_part), -y
        print(f"For x = {x}, the best y = {best_y} gives the maximum alternating part length:
{max_alternating_length}")

if __name__ == "__main__":
    main()
```

Output:

For x = 1, the best negative integer y = -1 gives the maximum alternating part length: 1
For x = 2, the best negative integer y = -1 gives the maximum alternating part length: 2
For x = 3, the best negative integer y = -2 gives the maximum alternating part length: 3
For x = 4, the best negative integer y = -3 gives the maximum alternating part length: 3
……
For x = 100, the best negative integer y = -62 gives the maximum alternating part length: 7

Q.2

The tail of a giant hare is attached by a giant rubber band to a stake in the ground. A flea is sitting on top of the stake eyeing the hare (hungrily). Seeing the flea, the hare leaps into the air and lands one kilometer from the stake (with its tail still attached to the stake by the rubber band). The flea does not give up the chase but leaps into the air and lands on the stretched rubber band one centimeter from the stake. The giant hare, seeing this, again leaps into the air and lands another kilometer from the stake (i.e., a total of two kilometers from the stake). The flea is undaunted and leaps into the air again, landing on the rubber band one centimeter further along. Once again the giant hare jumps another kilometer. The flea again leaps bravely into the air and lands another centimeter along the rubber band. If this continues indefinitely, will the flea ever catch the hare? (Assume the earth is flat and continues indefinitely in all directions.)

Solution: -

Let the number of jumps of the hare be denoted by n. After n jumps, the hare's total distance from the stake is:

Hare's distance after n jumps=100,000 × n cm

Let n be the number of jumps of the flea. The flea's total distance after n jumps is the sum of the first n integers, given by:

Flea's total distance after n jumps= $\frac{n(n+1)}{2}$cm

We want to determine the smallest n such that the flea's total distance is greater than or equal to the hare's total distance, i.e.,

$$\frac{n(n+1)}{2} \geq 100,000 \times n$$

Solving the Inequality:

This inequality holds for n≥199,999

Conclusion:

Therefore, the flea will catch the hare after **199,999 jumps**, as this is the smallest n for which the flea's total distance exceeds or equals the hare's total distance.

- This solution assumes that the flea's jumps grow in a simple arithmetic progression, and that the hare's jumps remain constant.
- The flea's growing jump length allows it to eventually surpass the hare, even though initially the hare leads by a large margin.

Q.3

The set S contains some real numbers, according to the following three rules.

(i) $\frac{1}{1}$ is in S.

(ii) If $\frac{a}{b}$ is in S, where $\frac{a}{b}$ is written in lowest terms (that is, a and b have highest common factor 1), then $\frac{b}{2a}$ is in S.

(iii) If $\frac{a}{b}$ and $\frac{c}{d}$ are in S, where they are written in lowest terms, then $\frac{a+c}{b+d}$ is in S.

These rules are exhaustive: if these rules do not imply that a number is in S, then that number is not in S. Can you describe which numbers are in S? For example, by (i), $\frac{1}{1}$ is in S. By (ii), since $\frac{1}{1}$ is in S, $\frac{1}{2\cdot 1}$ is in S. Since both $\frac{1}{1}$ and $\frac{1}{2}$ are in S, (iii) tells us $\frac{1+1}{1+2}$ is in S.

Solution: -

- Rule (i) tells us that $\frac{1}{1}$ (which is 1) is in SSS.
- Rule (ii) states that if $\frac{a}{b}$ is in SSS and gcd(a,b)=1 (i.e., a and b are coprime), then $\frac{b}{2a}$ is also in S.

For example, since $\frac{1}{1}\in S$, applying Rule (ii) gives us:

$$\frac{1}{2\times 1}\in S.$$

Next, since $\frac{1}{2}\in S$, applying Rule (ii) again:

$\frac{2}{2\times 1}$ (but this was already given by Rule (i))

From this, it's clear that we can keep creating fractions with powers of 2 in the numerator and denominator.

- Rule (iii) tells us that if $\frac{a}{b}$ and $\frac{c}{d}$ are in S, then $\frac{a+c}{b+d}$ is also in S, provided the fractions are in lowest terms.

We already know $\frac{1}{1}$ and $\frac{1}{2}$ are in S, so applying Rule (iii) gives us:

$\frac{2}{3}\in S$

Now, we have three numbers in S: $\frac{1}{1}, \frac{1}{2}, \frac{2}{3}$.

We can continue applying Rule (iii) with different combinations of these numbers:

- $\frac{1}{1}$ and $\frac{2}{3}$:

$\frac{3}{4} \in S$

- $\frac{1}{2}$ and $\frac{2}{3}$:

$\frac{3}{5} \in S$

Observations:

1. From Rule (ii), we can generate fractions of the form $\frac{1}{2^n}$, where n is a positive integer. This means that all fractions with a power of 2 in the denominator can be generated.
2. Rule (iii) allows us to combine any two fractions and create new fractions. This means that any fraction whose numerator and denominator are integers and whose greatest common divisor is 1 can eventually be formed by combinations of the numbers already in S.

Conclusion:

The set S consists of all fractions of the form $\frac{a}{b}$, where a and b are positive integers, and a and b are coprime (i.e., their greatest common divisor is 1). Thus, S includes all positive rational numbers.

Q.4

The sequence (x_n) of positive real numbers satisfies the relationship $x_{n-1}x_nx_{n+1} = 1$ for all $n \geqslant 2$. If $x_1 = 1$ and $x_2 = 2$, what are the values of the next few terms? What can you say about the sequence? What happens for other starting values?

The sequence (y_n) satisfies the relationship $y_{n-1}y_{n+1} + y_n = 1$ for all $n \geqslant 2$. If $y_1 = 1$ and $y_2 = 2$, what are the values of the next few terms? What can you say about the sequence? What happens for other starting values?

__Solution: -__

By using the recurrence relation to find the value of first few terms:

$x_3 = \frac{1}{2}$, $x_4 = 1$, $x_5 = 2$

The sequence alternates in a periodic manner, following the pattern:

$x_1 = 1$, $x_2 = 2$, $x_3 = \frac{1}{2}$, $x_4 = 1$, $x_5 = 2$, ….

The sequence repeats every 3 terms, so it is periodic with a period of 3.

With this we can find for any value of x_n .

- If n is any multiple of 3, $x_n = \frac{1}{2}$
- If n is 3a + 1 (for some whole number a), $x_n = 1$
- If n is 3a + 2 (for some whole number a), $x_n = 2$

With other starting value it still follows the pattern of periodic manner with period 3.

Using the recurrence relation to find the values of first few terms:

$y_3 = -1$, $y_4 = 1$, $y_5 = 0$, $y_6 = 1$, $y_7 = 1$, $y_8 = 0$, $y_9 = 1$, $y_{10} = 1$

Thus, the sequence seems to repeat with a cycle of length 6.

For n=6

$y_5y_7 + y_6 = 1$

Substitute $y_5 = 0$ and $y_6 = 1$:

$0 \cdot y_7 + 1 = 1 \quad \Rightarrow \quad 1 = 1$

The sequence is alternating in a particular pattern which is true, but does not provide information on y_7. Since the term involving y_5 vanishes, we have no new constraint on y_7. Based on this behavior, we can hypothesize that the sequence might repeat with a cycle length of 6.

Thus, the next few terms will likely follow the same pattern, with a periodic repetition emerging.

Let's examine how changing the initial conditions might affect the sequence.

Starting with $\mathbf{y_1}=2$ and $\mathbf{y_2}=3$:

Using the same recurrence relation, we can find the next few terms by following the same steps as before:

For n=2:

$y_1 y_3 + y_2 = 1 \quad \Rightarrow \quad 2 \cdot y_3 + 3 = 1 \quad \Rightarrow \quad 2y_3 = -2 \;\Rightarrow \quad y3 = -1.$

Generally, the behavior of the sequence depends on the starting values, but the recurrence relation typically leads to sequences that either have periodic behavior or specific growth patterns depending on the values chosen.

For the initial values $y_1=1$ and $y_2=2$, the sequence seems to follow a periodic pattern. However, the behavior may differ significantly for different initial values, and further investigation into specific cases is needed to determine how the sequence evolves with other starting conditions.

Q.5

A monkey has filled in a 3 × 3 grid with the numbers 1, 2, . . . , 9. A cat writes down the three numbers obtained by multiplying the numbers in each horizontal row. A dog writes down the three numbers obtained by multiplying the numbers in each vertical column. Can the monkey fill in the grid in such a way that the cat and dog obtain the same lists of three numbers? What if the monkey writes the numbers 1, 2, . . . , 25 in a 5 × 5 grid? Or 1, 2, . . . , 121 in a 11 × 11 grid? Can you find any conditions on n that guarantee that it is possible or any conditions that guarantee that it is impossible for the monkey to write the numbers 1, 2, . . . , n^2 in an n × n grid so that the cat and the dog obtain the same lists of numbers?

Solution: -

We begin by demonstrating that for small grids (such as n=3), it is indeed possible for the monkey to fill the grid in such a way that the products of the rows match the products of the columns, in any order.

Consider the following grid:

5 3 2 6 7 4 1 8 9

- **Row products (Cat's list)**:
  - Row 1: 5×3×2=30
  - Row 2: 6×7×4=168
  - Row 3: 1×8×9=72

  So, the cat's products are: 30,168,72

- **Column products (Dog's list)**:
  - Column 1: 5×6×1=30
  - Column 2: 3×7×8=168
  - Column 3: 2×4×9=72

  So, the dog's products are also: 30,168,72

Since the cat and dog both obtain the same list of products, just in different orders, this confirms that for the 3×3 grid, **it is possible** for the monkey to arrange the numbers such that the row and column products match (in any order).

Generalizing for Larger Grids

Now let's consider larger grids, specifically for an n×n grid where the numbers 1,2,...n^2 must be placed in the cells.

For large grids, a significant challenge arises due to the distribution of **prime numbers**. Prime numbers limit the flexibility in arranging the numbers such that the products of the rows and columns can match.

The Role of Prime Numbers

If we place primes in rows or columns, the products of those rows or columns are "fixed" to some extent, as they depend directly on the prime numbers. For example, consider the prime numbers in the range ($\frac{n(n)}{2}$,n^2). Placing prime in rows or columns limits the ways in which the other numbers can adjust the product. This can prevent the row products from matching the column products.

Number of Primes in the Range ($\frac{n(n)}{2}$,n^2)

For a general n×n grid, the number of primes in the range ($\frac{n(n)}{2}$,n^2) grows as n increases. This can be a critical factor in determining whether it is possible to arrange the numbers so that the row and column products match.

Conclusion

- **For small grids** (like n=3), the monkey can indeed arrange the numbers such that the products of the rows and columns match in any order.
- **For large grids**, especially when n is large enough that the number of primes in the range ($\frac{n(n)}{2}$,n^2) is substantial, it becomes **impossible** for the monkey to arrange the numbers in a way that satisfies the condition.

Reference

This was a question previously asked in PROMYS 2023, which I tried to solve so the matrix and the conjecture which I explained is taken from my previous solution.

Q.6

Consider the polynomial $(x + y + z)^{n}$ for n ≥ 1. Expand and combine like terms so that all terms of the form $x^i y^j z^k$ are together with one coefficient. Let c_n(i, j, k) be this coefficient for fixed n.

(i) Can you predict when c_n(i, j, k) is even or odd? For example, when n = 1, we have c_1(0, 0, 1), c_1(0, 1, 0), and c_1(0, 0, 1) are all equal to 1, hence odd, and not defined for any other values of i, j, k.

(ii) Can you say anything about the sum of the coefficients c_n(i, j, k) over the values where i, j, k > 0? Can you find an expression for this sum?

Solution: -

The multinomial coefficient c_n(i, j, k) = $n(i\,j\,k)$ determines the coefficients in the expansion of $(x + y + z)^{n}$.

- For n=1, the expansion of $(x + y + z)^{n} = x + y + z$ gives:
  c_1(0, 0, 1), c_1(0, 1, 0), c_1(0, 0, 1)

These are all odd and no other terms exist because i + j + k = 1 and the only valid combinations are (1, 0, 0), (0,1,0) and (0, 0, 1).

- For n = 2, the expansion of $(x + y + z)^{2} = x^2 + y^2 + z^2$+ 2xy + 2xz + 2yz gives:
- c_2(2, 0, 0) = 1, c_2(0, 2, 0) = 1, c_2(0, 0, 2) = 1
- c_2(1, 1, 0) = 2, c_2(1, 0, 1) = 2, c_2(0, 1, 1) = 2

The coefficients are either odd (1) or even (2). In this case, terms where two of the exponents are 1 have even coefficients, while the terms with a single exponent of 2 have odd coefficients.

The total sum of coefficients in $(x + y + z)^{n}$ is 3^{n}, and the sum where i, j, k > 0 can be found using inclusion-exclusion.

For n = 2, we already know that the total sum of coefficients in $(x + y + z)^{2}$ is:

$3^2 = 9$

Now, let's compute the sum where i, j, k > 0:

- The full expansion of $(x + y + z)^2$ is:
  $(x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2xz + 2yz$
  The total sum is 1 + 1 + 1 + 2 + 2 + 2 = 9.

- Now, we exclude the terms where one or more of i, j, or k is zero:

  - There are 3 terms where exactly one exponent is zero: x^2, y^2 and z^2.
  - We exclude these terms from the total, leaving:
- 9 - 3 = 6

Thus, the sum of the coefficients where i, j, k > 0 is 6.

There are 3 variables (x, y, and z), so the total number of terms where one of the exponents is zero is:

$3.(n - 1)$

- The total sum of all coefficients is 3^n
- Number of terms where one exponent is zero, which is $3.(n - 1)$
- Number of terms where two exponents are zero, which is 3

$$S = 3^n - 3.(n - 1) + 3$$

This gives us:

$$S = 3^n - 3n + 6$$

For n = 2, this gives:

$S = 3^n - 3n + 6$ = 9 - 9 + 6 = 6

Thus, the sum of coefficients where i, j, k > 0 is 6 for n = 2.

To cross-verify the validity of this derived formula and for further analysis. I have made a python program to loop values of n from 1 to any number.

Code:

```
def multinomial_sum(n):
    return 3**n - 3 * (n - 1) + 3
def check_formula():
    for n in range(1, 101):
        if multinomial_sum(n) != 3**n - 3*n + 6:
            print(f"Mismatch for n = {n}")
        else:
            print(f"Valid for n = {n}")
check_formula()
```

Output:

```
Valid for n = 1
Valid for n = 2
Valid for n = 3
Valid for n = 4
...
Valid for n = 100
```

Q.7
Can you find an equilateral triangle in the xy-plane with all its vertices having integer coordinates? If so, give an example, and if not, explain why no such triangle exists. Can you find an equilateral triangle with two vertices that have integer coordinates while the third vertex is within $\frac{1}{10}$ distance of a point with integer coordinates? What about within $\frac{1}{100}$? Within $\frac{1}{n}$?

<u>Solution: -</u>

We are looking for an equilateral triangle where the coordinates of all three vertices are integers.

The distance between two points with integer coordinates is given by the distance formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

The square roots involved lead to irrational numbers for distances, which cannot coincide with distances that would arise from integer coordinates in an equilateral triangle.

Hence, it is impossible to construct an equilateral triangle with all integer coordinates.

Assume two of the vertices, say A(x_1, y_1) and B(x_2, y_2) have integer coordinates.

The third vertex C$(x_3, y_3)$is constrained to lie on a circle centered at the midpoint of A and B, with radius equal to the side length of the equilateral triangle.

This circle is the locus of points that are equidistant from A and B. Since the coordinates of A and B are integers, the midpoint of A and B has rational coordinates, and the radius is a rational number.

Due to the geometric constraints, the third vertex C will generally not lie on a lattice point. But it can be very close to a lattice point.

As we decrease the required distance (say, to $\frac{1}{100}$ or $\frac{1}{n}$), the third vertex can be made close to an integer point by choosing an appropriate side length for the equilateral triangle.We can always find a position for the third vertex that is close to an integer lattice point.

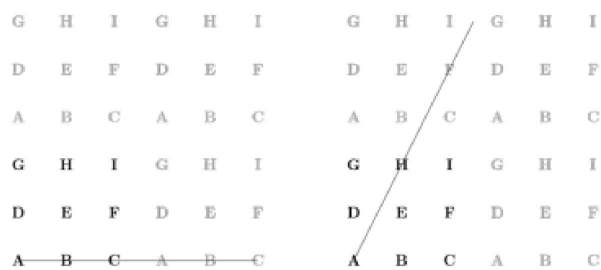
Thus, for any small distance, there exists a configuration of the equilateral triangle such that the third vertex is within that distance of an integer lattice point. The vertex cannot exactly coincide with an integer point.

Q.8

Consider the grid of letters that represent points below, which is formed by repeating the bold 3×3 grid. (The points are labeled by the letters A through I.)

G	H	I	G	H	I
D	E	F	D	E	F
A	B	C	A	B	C
G	**H**	**I**	G	H	I
D	**E**	**F**	D	E	F
A	**B**	**C**	A	B	C

We'll draw lines through the bottom left point (the bold A) and at least one other bold letter, then see what set of letters the line hits. We've drawn two example lines for the repeating 3×3 grid below. For the example on the left, the set of letters is $\{A, B, C\}$, and for the right, the set of letters is $\{A, F, H\}$.



Assuming the 3×3 grid repeats forever in every direction, do any of these lines ever pass through more than 3 different letters? Can you get the same set of letters from two different lines? Find the four different sets of letters that you can get from drawing lines in this grid.

What would happen if you had a repeated 5×5 grid of letters (and still had to draw lines through the bottom left point and at least one other bold point)? Can you predict what would happen with a repeated 7×7 grid? Does your prediction also work for 6×6? Can you justify your predictions?

Solution: -

When you draw a straight line from one point to another in the grid, the line will not just connect two points on the same 3 × 3 block — it can also pass through points in adjacent blocks in the repeating grid.

The challenge here is to draw a straight line from A to another point, which can pass through multiple blocks as it extends. The line will hit a series of letters from the repeating pattern, and the task is to find out which letters the line intersects.

Example: Drawing a Straight Line from A to H

Let’s revisit the idea of drawing a straight line from A to H:

- A is located at (3,1) in the first 3 × 3 block (bottom-left corner).
- H is located at (1,2) in the first 3 × 3 block (top-center).

Now, if we draw a straight line from A to H, this line will indeed pass through more than just A and H in the first 3 × 3 block. The line will also hit the letter F in the second 3 × 3 block in the top-right corner.

Here’s what happens:

1. The line starts at A (bottom-left of the first block).
2. It passes through B and C (along the first block).
3. It continues into the next block and passes through F in the top-right corner.
4. It eventually passes through H in the original block, which is located at the top-center.

Thus, the line will pass through A, F, H.

So, the set of letters this line passes through is:

- {A, F, H}

Key Observations:

- When you draw a straight line from A to H, it passes through A, F, and H because the straight line extends into adjacent blocks.

Let’s now reconsider the set of letters for any line:

- A to B and C: This line only stays within the original block, passing through A, B, and C.
- A to E and H: This line will pass through A, and then E and H, but it crosses into neighboring blocks depending on the direction.
- A to F and I: As with the other cases, the line will cross into neighboring blocks, passing through A, F, and I.

Conclusion:

- A straight line from A to H actually passes through A, F, and H — because it intersects the top-right block at F before reaching H in the original block.

This creates a more complex pattern of sets of letters, and no line will pass through more than 3 distinct letters from any repeating 3 × 3 block, but it can hit 3 different letters in several blocks along its straight path.